

Signal and Systems

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Energy of a Signal

If $x(t)$ represents a signal, then the energy of that signal can be represented as:

$$E_x = \int_{-\infty}^{\infty} x^2(t) dt$$

For a complex-valued signal $x(t)$:

$$E_x = \int_{-\infty}^{\infty} |x(t)|^2 dt$$

Units: If $x(t)$ is Voltage: $V^2 \cdot s$ If $x(t)$ is Current: $A^2 \cdot s$

If the amplitude of $x(t)$ tends to 0 as $t \rightarrow \infty$, then the signal has a finite amount of energy.

Power of a Signal

The average power of the signal is given by:

$$P_x = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} |x(t)|^2 dt$$

Unit: V^2 or A^2

$|x(t)|$ is used when $x(t)$ is complex-valued. $x(t)^2$ is used when $x(t)$ is real-valued.

RMS and Power of Signals

$$\text{RMS} = \sqrt{P_x}$$

Power is generally calculated for a signal with the same value period (T) or having some statistical regularity.

Examples

Ramp signal: For a ramp signal

$$x(t) = t$$

it is neither having finite energy nor finite power.

Unit step function: For a unit step function

$$x(t) = u(t)$$

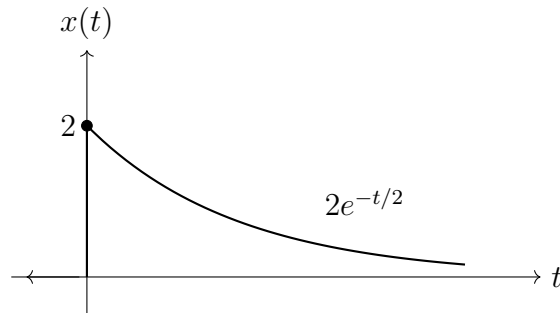
it is neither periodic nor statistically regular. It has finite power:

$$u(t) = \begin{cases} 1, & t \geq 0, \\ 0, & t < 0. \end{cases}$$

Example Signal

Consider the signal

$$x(t) = \begin{cases} 2, & t = 0, \\ 2e^{-t/2}, & t > 0. \end{cases}$$



Amplitude of $x(t) \rightarrow 0$ as $t \rightarrow \infty$. Energy and Power Calculations
The energy of a signal is given by:

$$E_x = \int_{-\infty}^{\infty} x^2(t) dt$$

For the given signal:

$$E_x = \int_{-\infty}^{-1} (0)^2 dt + \int_{-1}^0 (2)^2 dt + \int_0^{\infty} (2e^{-t/2})^2 dt$$

Calculating each term:

$$\int_{-1}^0 4 dt = 4[t]_{-1}^0 = 4(0 - (-1)) = 4$$

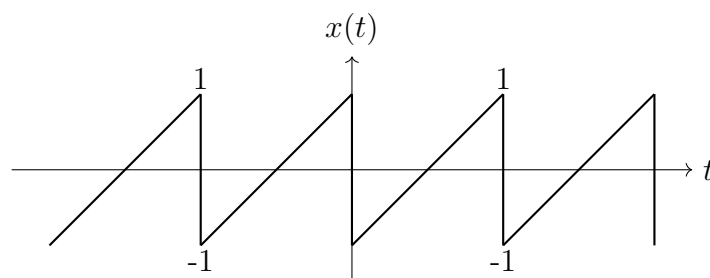
$$\int_0^{\infty} 4e^{-t} dt = 4[-e^{-t}]_0^{\infty} = 4(0 - (-1)) = 4$$

Therefore:

$$E_x = 0 + 4 + 4 = 8$$

Example 2: Sawtooth Signal

$x(t) = \left\{ \begin{array}{l} \text{Linear ramp from } -1 \text{ to } 1 \text{ over period } T = 2, \\ \text{periodic} \end{array} \right.$



Slope Calculation

From the equation of a straight line:

$$y = mx$$

For the ramp section:

$$x(t) = mt$$

Given that at $t = 1$, $x(t) = 1$:

$$1 = m(1) \quad \Rightarrow \quad m = 1$$

Power of the Sawtooth Signal

Power is given by:

$$P_x = \frac{1}{T} \int_{-T/2}^{T/2} x^2(t) dt$$

Here, $T = 2$:

$$\begin{aligned} P_x &= \frac{1}{2} \int_{-1}^1 t^2 dt \\ &= \frac{1}{2} \left[\frac{t^3}{3} \right]_{-1}^1 = \frac{1}{2} \left(\frac{1}{3} - \left(-\frac{1}{3} \right) \right) = \frac{1}{2} \cdot \frac{2}{3} = \frac{1}{3} \end{aligned}$$

Example: Power and RMS of a Cosine Signal

Given:

$$x(t) = C \cos(\omega_0 t + \theta)$$

The power is:

$$P_x = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} x^2(t) dt$$

Substitute $x(t)$:

$$P_x = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} C^2 \cos^2(\omega_0 t + \theta) dt$$

Using the identity $\cos^2 \alpha = \frac{1 + \cos 2\alpha}{2}$:

$$\begin{aligned} P_x &= \frac{C^2}{T} \int_{-T/2}^{T/2} \frac{1 + \cos(2\omega_0 t + 2\theta)}{2} dt \\ &= \frac{C^2}{2T} \int_{-T/2}^{T/2} [1 + \cos(2\omega_0 t + 2\theta)] dt \end{aligned}$$

Splitting the integrals:

$$P_x = \frac{C^2}{2T} \int_{-T/2}^{T/2} 1 dt + \frac{C^2}{2T} \int_{-T/2}^{T/2} \cos(2\omega_0 t + 2\theta) dt$$

First term:

$$\frac{C^2}{2T} \cdot T = \frac{C^2}{2}$$

Second term:

$$\frac{C^2}{2T} \cdot \left[\frac{\sin(2\omega_0 t + 2\theta)}{2\omega_0} \right]_{-T/2}^{T/2} \xrightarrow{T \rightarrow \infty} 0$$

Therefore:

$$P_x = \frac{C^2}{2}$$

The RMS value is:

$$\text{RMS} = \sqrt{P_x} = \sqrt{\frac{C^2}{2}} = \frac{C}{\sqrt{2}}$$

Example

Given:

$$x(t) = D e^{j\omega t}$$

The power is:

$$P_x = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} |D e^{j\omega t}|^2 dt$$

Note:

$$|e^{j\theta}| = 1, \quad e^{j\theta} = \cos \theta + j \sin \theta$$

Thus:

$$|e^{j\omega t}|^2 = 1$$

So:

$$P_x = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} D^2 \cdot 1 dt = \lim_{T \rightarrow \infty} \frac{D^2}{T} \cdot T = D^2$$

Therefore:

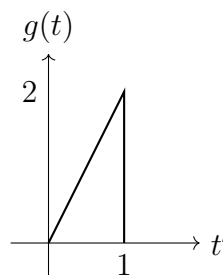
$$\text{RMS} = \sqrt{P_x} = \sqrt{D^2} = D$$

Example

Let:

$$g(t) = \begin{cases} 2t, & 0 \leq t \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

Here slope $m = 2$.



Energy of $g(t)$:

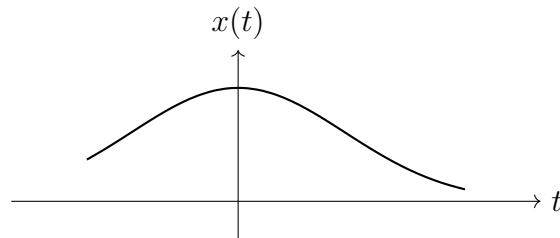
$$E_g = \int_{-\infty}^{\infty} g^2(t) dt$$

Since $g(t)$ is nonzero only from 0 to 1:

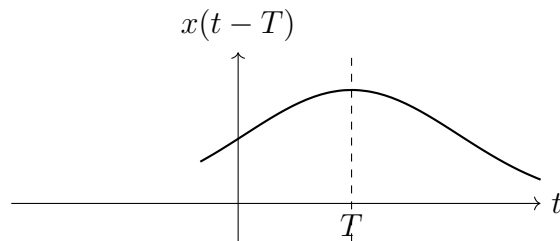
$$\begin{aligned} E_g &= \int_0^1 (2t)^2 dt = 4 \int_0^1 t^2 dt \\ &= 4 \left[\frac{t^3}{3} \right]_0^1 = \frac{4}{3} \end{aligned}$$

Signal Operation

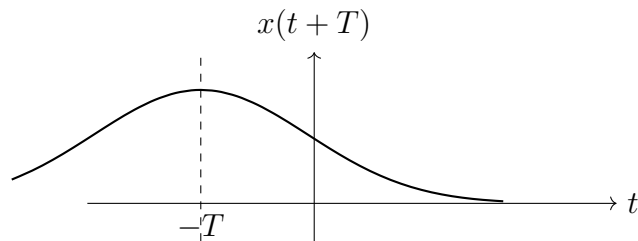
Original Signal $x(t)$



Time Delay: $x(t - T)$

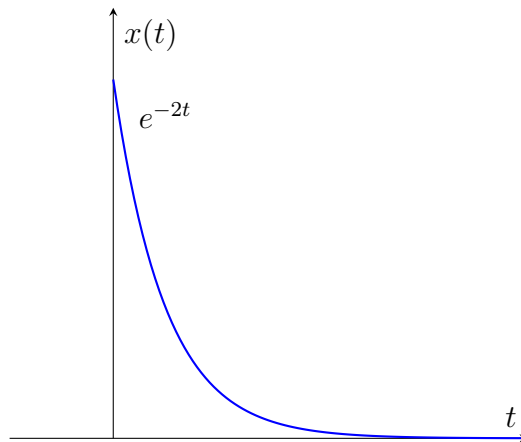


Time Advance: $x(t + T)$

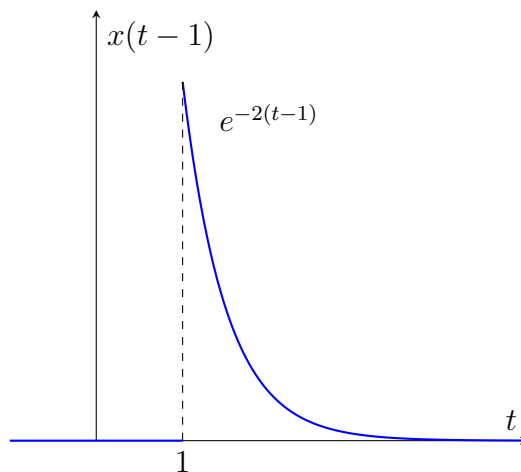


Signal Shifting Example

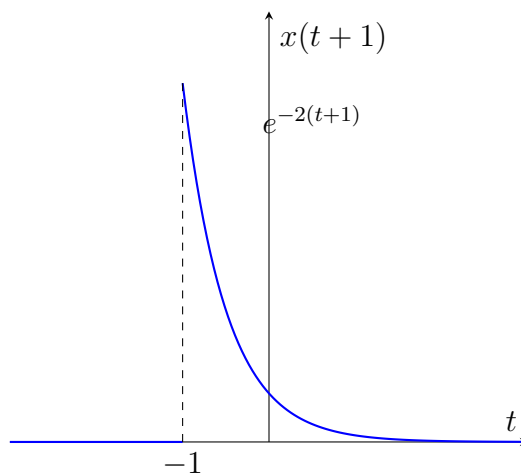
Original Signal: $x(t) = e^{-2t}$



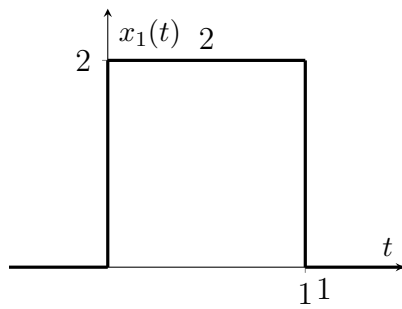
Signal Delayed by 1 second (Shift Right): $x(t - 1)$



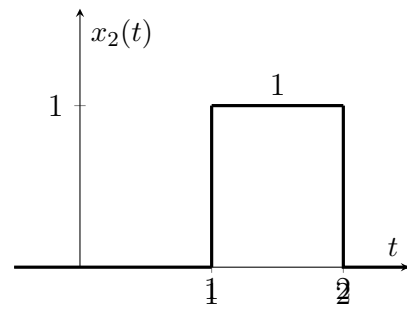
Signal Advanced by 1 second (Shift Left): $x(t + 1)$



Signals and Their Energies

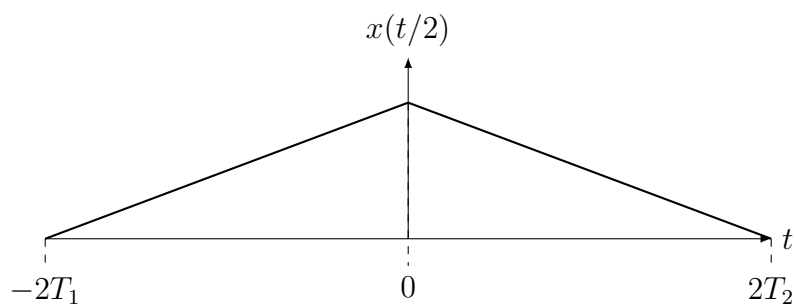
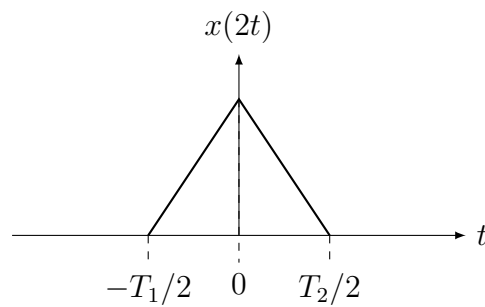
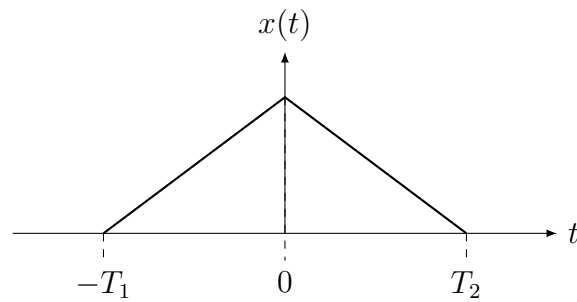


$$E_{x_1} = \int_0^1 2^2 dt = 4$$



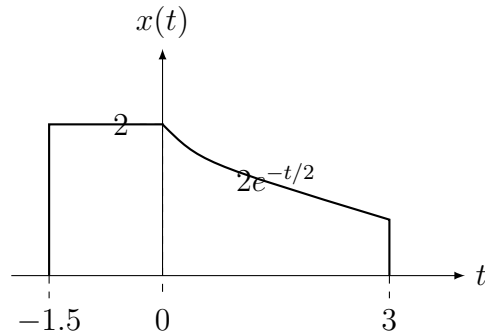
$$E_{x_2} = \int_1^2 1^2 dt = 1$$

Time Scaling



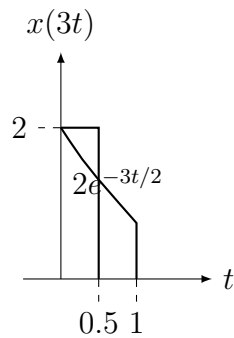
- $x(at)$ represents **compression** if $a > 1$
 - $x\left(\frac{t}{a}\right)$ represents **expansion** if $a > 1$
 - $x(at)$ represents **expansion** if $a < 1$
- $x\left(\frac{t}{a}\right)$ represents **compression** if $a < 1$ Time Scaling Example

1) Original Signal $x(t)$



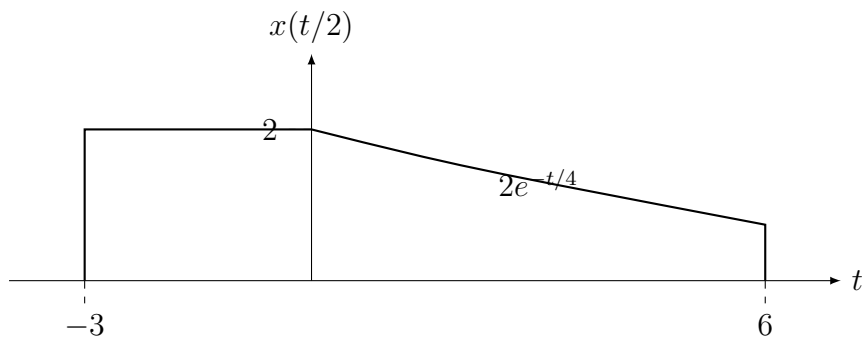
$$x(t) = \begin{cases} 2e^{-t/2}, & 0 \leq t < 3 \\ 2, & -1.5 \leq t < 0 \\ 0, & \text{otherwise} \end{cases}$$

*2) Compression: $x(3t)$



$$x(3t) = \begin{cases} 2e^{-3t/2}, & 0 \leq t < 1 \\ 2, & 0.5 \leq t < 0 \\ 0, & \text{otherwise} \end{cases}$$

3) Expansion: $x(t/2)$



$$x(t/2) = \begin{cases} 2e^{-t/4}, & 0 \leq t \leq 6 \\ 2, & -3 \leq t < 0 \\ 0, & \text{otherwise} \end{cases}$$

Question

Show that if $\omega_1 = \omega_2$, the power of
 $x(t) = C_1 \cos(\omega_1 t + \theta_1) + C_2 \cos(\omega_2 t + \theta_2)$
 is

$$C_1^2 + C_2^2 + 2C_1 C_2 \cos\left(\frac{\theta_1 - \theta_2}{2}\right)$$

Solution

Given:

$$x(t) = C_1 \cos(\omega_1 t + \theta_1) + C_2 \cos(\omega_2 t + \theta_2)$$

For $\omega_1 = \omega_2 = \omega$, we have:

$$x(t)^2 = C_1^2 \cos^2(\omega t + \theta_1) + C_2^2 \cos^2(\omega t + \theta_2) + 2C_1 C_2 \cos(\omega t + \theta_1) \cos(\omega t + \theta_2)$$

Power is:

$$P = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} x(t)^2 dt$$

Substitute:

$$P = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} [C_1^2 \cos^2(\omega t + \theta_1) + C_2^2 \cos^2(\omega t + \theta_2) + 2C_1 C_2 \cos(\omega t + \theta_1) \cos(\omega t + \theta_2)] dt$$

We know:

$$\frac{1}{T} \int_{-T/2}^{T/2} \cos^2(\omega t + \phi) dt = \frac{1}{2}$$

and

$$\cos A \cos B = \frac{\cos(A - B) + \cos(A + B)}{2}$$

So:

$$P = \frac{C_1^2}{2} + \frac{C_2^2}{2} + \frac{1}{T} \int_{-T/2}^{T/2} C_1 C_2 [\cos(\theta_1 - \theta_2) + \cos(2\omega t + \theta_1 + \theta_2)] dt$$

The second cosine term integrates to zero over a symmetric infinite interval, so:

$$P = \frac{C_1^2}{2} + \frac{C_2^2}{2} + C_1 C_2 \cos(\theta_1 - \theta_2)$$

Finally:

$$P = \frac{C_1^2 + C_2^2 + 2C_1 C_2 \cos(\theta_1 - \theta_2)}{2}$$